

BLUEPRINTIQ TECHNICAL MEMORANDUM

UNCERTAINTY-AWARE BLUEPRINT-TO-BOQ INTELLIGENCE | DETAILED ANALYSIS REPORT

STATUTORY ENGINEERING NOTICE: BlueprintIQ provides automated preliminary computer-vision geometry extraction and material estimations. This document does not replace a licensed professional engineer, architect, quantity surveyor, or municipal authority approval. All quantities must be field-verified before procurement.

1. PROJECT & DRAWING METADATA

Project Name:	Villa Residence G+1	Building Type:	Residential
Floors:	2	Unit System:	METRIC
Target Area:	180.0 m²	Soil Type:	Not Provided
Drawing Scale:	1:100	Concrete Grade:	M20
Wall Thickness:	230 mm	Overall Confidence:	MEDIUM
Overall Confidence Score:	87% (Estimated AI/CV Analysis)	Estimation Basis:	Computer-vision floor-plan extraction & quantity modeling

2. CONSOLIDATED MATERIAL ESTIMATES (UNCERTAINTY-AWARE)

MATERIAL	ESTIMATED	RANGE	CONF.	CALCULATION BASIS
Cement (OPC/PPC 50kg Bags)	2,152.0 bags	1,980 - 2,367	MEDIUM	Concrete (557.4 bags) + Masonry (1019.8 bags) + Plaster (574.6 bags)
River Sand / M-Sand	336.8 m³	303 - 377	MEDIUM	Concrete (29.0m³) + Masonry (212.5m³) + Plaster (95.3m³)
Coarse Aggregate (20mm & 10mm Graded)	84.8 m³	76 - 93	MEDIUM	RCC Concrete (58.1m³) + PCC Bed (26.8m³)
Modular Clay / Fly Ash Bricks	296,499.0 nos	272,779 - 326,149	HIGH	564.76 m³ net masonry x 500 bricks/m³ + 5% breakage allowance
Reinforcement Steel (Fe500 TMT Rebars)	5,172.0 kg	4,396 - 6,464	MEDIUM	Slabs (3391.0 kg at 0.8% vol) + Beams/Columns (1780.0 kg at 1.5% vol)
Vitrified Floor Tiles (600x600mm)	34.7 m²	33 - 38	HIGH	Net floor area 32.1m² + 8% cutting and breakage allowance
Emulsion Paint & Primer (2 Coats)	564.6 litres	508 - 649	MEDIUM	Total plaster & ceiling area 5645.9m² / coverage 10 m²/Litre

3. DETAILED BILL OF QUANTITIES (CIVIL & FINISHES)

CODE	DESCRIPTION	QTY	UNIT	RANGE	CONF.
CIV-01	Earthwork excavation in foundation trenches	434.52	m³	304.2 - 630.0	LOW
CIV-02	Plain Cement Concrete (PCC 1:4:8) in foundation bed (100mm)	31.49	m³	28.3 - 36.2	MEDIUM
CIV-03	Reinforced Cement Concrete (M20) for Slabs, Beams, Columns	69.12	m³	63.6 - 77.4	MEDIUM
CIV-04	Brick Masonry in Cement Mortar (1:6) with Modular Bricks	564.76	m³	508.3 - 621.2	HIGH
CIV-05	Cement Plastering (12mm Internal 1:6 + 18mm External 1:4)	5285.90	m²	4915.9 - 5814.5	MEDIUM
CIV-06	Flooring with Vitrified Tiles including 8% wastage	34.67	m²	32.9 - 37.4	HIGH

4. BLUEPRINT ANOMALIES & CODE COMPLIANCE FINDINGS

ID	SEV.	FINDING & EVIDENCE	RECOMMENDED ACTION
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ISSUE-001	MEDIUM	Potential Habitable Area Constraint: LIVING & DINING Measured area: 3.8 m² vs Recommended threshold: 9.5 m² <i>Standard: NBC 2016 Part 3 Clause 4.2.1 (Minimum Habitable Room Area)</i>	Review room layout with a licensed architect to confirm local municipal development control guidelines. Status: Verification Required
ISSUE-002	MEDIUM	Staircase Riser and Tread Detail Missing on 2D Plan No textual riser/tread callouts (e.g. '18 Risers @ 165mm') identified in staircase zone. <i>Standard: IBC Section 1011.5.2 & NBC 4.8.1</i>	Request sectional drawing / stair detailing schedule for precise concrete and reinforcement calculation. Status: Verification Required

5. LOCAL AI (GEMMA 2B) SYNTHESIS & REASONING

Executive Synthesis: Concise engineering analysis of the drawing completeness and geometry

Uncertainty Assessment: Clear explanation of how missing parameters affect accuracy

6. MISSING PARAMETERS & ENGINEERING ASSUMPTIONS

- Missing Input Parameters:**
- **Soil Type:** Not Provided — *Foundation trench excavation, footing depth, and sub-base quantities cannot be verified without soil bearing capacity (SBC).*
- Applied Engineering Assumptions:**
- Excavation depth assumed at nominal 1.2m for medium dense soil; footing sizes subject to soil report.
 - Reinforcement steel estimated via standard IS 456 thumb rules (0.8% for slabs, 1.5% for framed elements).